

Ψ . The ocean researcher's workspace



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[GitHub Repository](#)

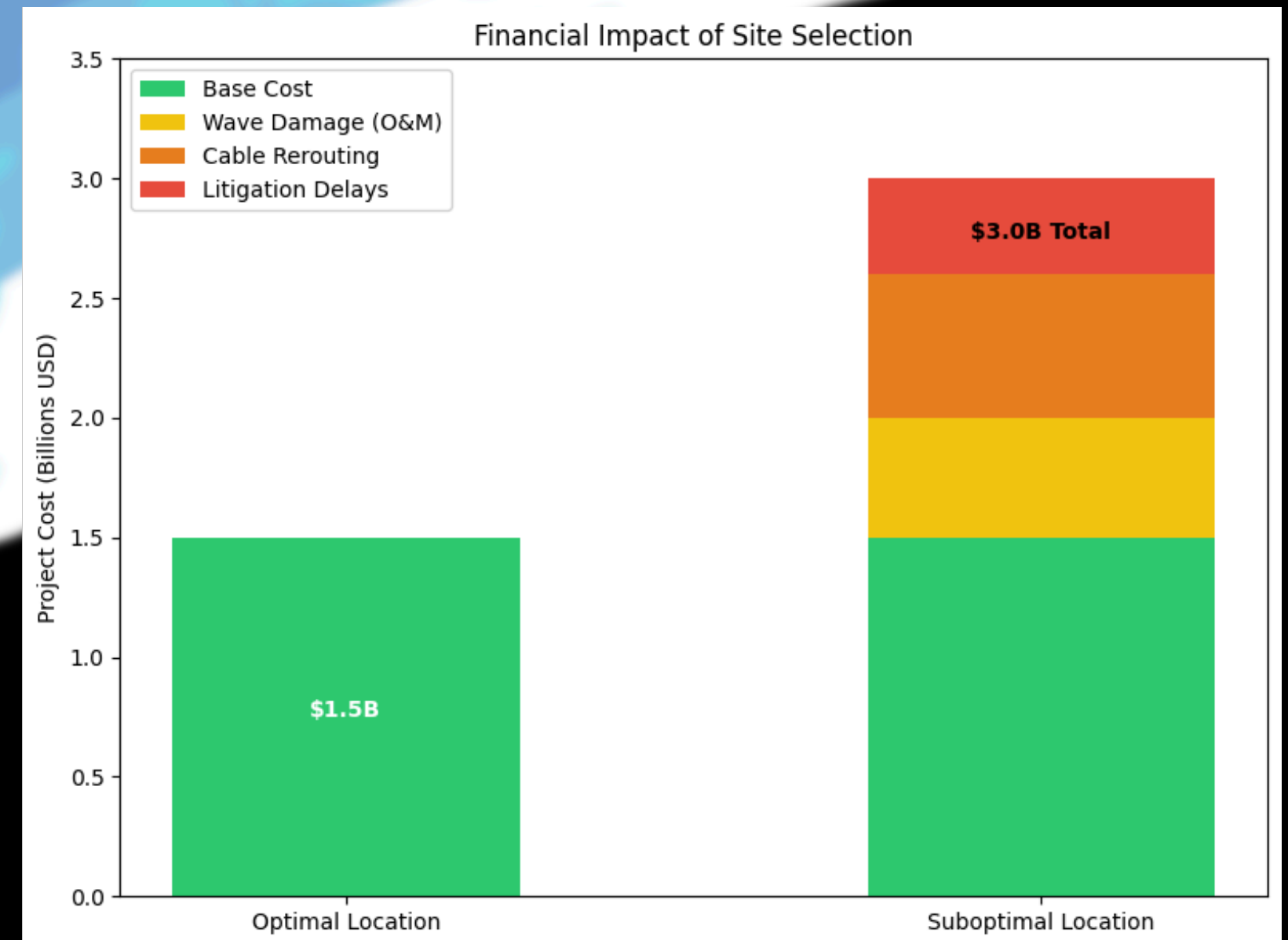
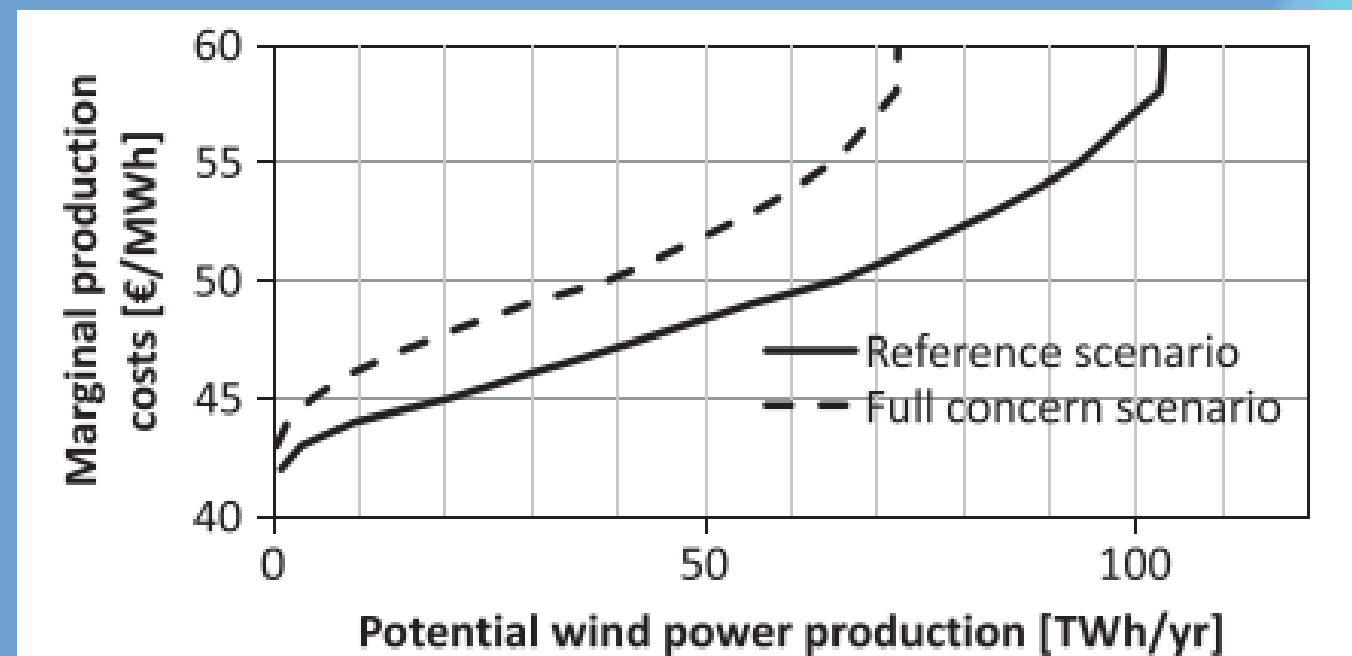
A BRIEF INTRODUCTION TO OFFSHORE WIND

- A single 15MW turbine can power up to 15,000 homes.
- Offshore wind harnesses the massive, consistent wind energy over the ocean.
- However, growing sustainable energy systems without disrupting delicate marine environments is a huge engineering challenge



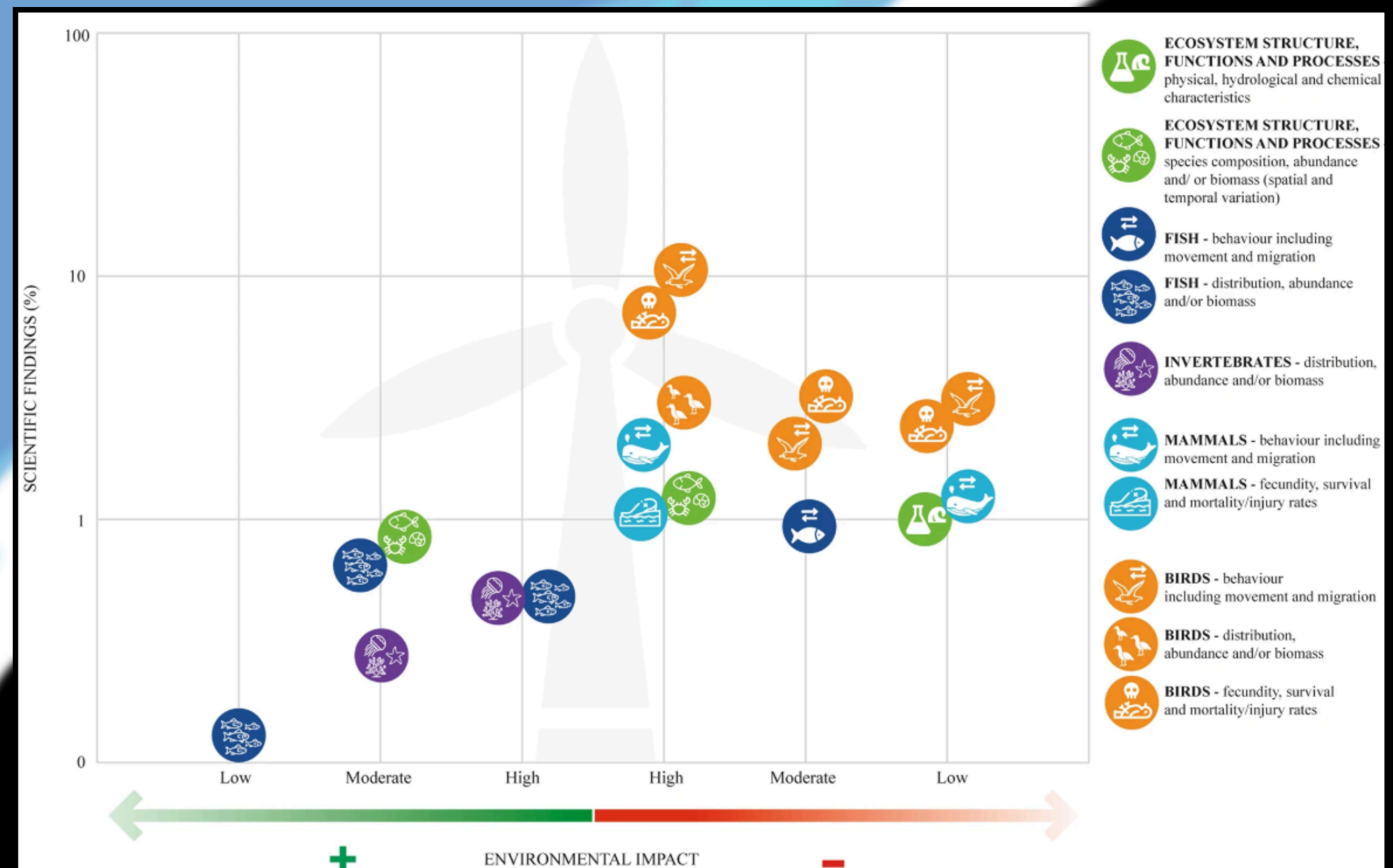
WHAT HAPPENS IF YOU PICK THE WRONG LOCATION?

- > O&M Cost Explosions: Unpredictable waves can drive lifetime costs up by 30%.
- > Geological Blindspots: Millions lost routing cables around unexpected seabed trenches.
- > Litigation Delays: Siting near protected habitats halts financing.



WHAT HAPPENS IF YOU PICK THE WRONG LOCATION?

- EMF disruptions to shark and ray navigation.
- Turbine collisions in seabird migratory corridors.
- The "Artificial Reef" Paradox: Foundations act as stepping stones for invasive species.



THE PROBLEM: Legacy Tools are too slow!

Marxan: Requires a geospatial data scientist and weeks of manual data prep to run heavy, academic optimizations.

SeaSketch: Forces users to manually draw boundaries and visually interpret overlapping, static map layers.

Government Databases (EMODnet): Provide raw map layers but zero instant analysis, creating an unreadable mess of colors.

To hit the 5GW by 2030 target, the NS government just spent *months* manually overlaying spatial constraints (25-km buffers, shallow geology, fishing zones) to identify Tier 1 areas like Sydney Bight and Sable Island Bank.

While tools like Marxan needs weeks of data prep, SeaSketch relies on human guesswork, and MarineCadastre just dumps raw layers on a map, PSI fuses 169 live variables, runs the physics automatically, and outputs a single defensible score in seconds.

WHAT Ψ DOES DIFFERENT:

PSI simulates the environmental impact of offshore wind turbine placement before anything hits the water, so developers can find sites that generate clean energy without destroying the marine ecosystems around them.

WHO THIS BENEFITS:

Researchers
2,900+

offshore wind publications per year
169 live variables, 16 real APIs, all in one platform. No more stitching together 6 different data portals per study.

Government & regulators
\$14.5M+

DOE spends per round of env. research
PSI delivers the same outputs — NEPA, ESA, DFO-ready, at a fraction of the cost, with full data provenance.

Investors & developers
1.4M

workers in the offshore wind sector
Risk-adjusted IRR, NPV, LCOE, all uncertainty-quantified via MCMC. Bankable numbers, not point estimates.

Cost Comparison

\$2–5M

Traditional EIA per offshore wind project

2–4 yrs

Typical regulatory approval timeline

Minutes

PSI simulation, same physics, real data, any site

WHAT Ψ DOES DIFFERENT:

PSI simulates the consequences of siting before you commit.

Place & configure

Drop a turbine pin on a 3D ocean globe. Set foundation type, rotor size, hub height, cable specs.

Real ocean data in

169 live variables, ERA5 winds, Copernicus currents, GEBCO seabed, AIS shipping, species records.

169

live ocean variables

Peer-reviewed physics run
Gaussian wake, François-Garrison acoustics, Soulsby scour, RK4 Lagrangian tracking, MaxEnt species habitat.

5

peer-reviewed physics model

What it answers?

Wake turbulence cutting energy yield?
Construction noise harming marine mammals?
Overlap with Right Whale or seabird corridors?
Where do spills disperse under real tidal conditions?
Defensible under NEPA, ESA & DFO review?
Scour risk given this seabed sediment?

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FEATURES CAPABILITIES:

PSI is built to expand

Any ocean, any region

Built on global APIs, Copernicus, ERA5, GEBCO, OBIS. Expanding from Scotian Shelf to North Sea, Gulf of Mexico, or Southeast Asia is a config change, not a rebuild.

Regulatory pre-screening tool

Direct submission to DFO, NEPA, and ESA portals already scoped. PSI outputs become the environmental pre-screening package, cutting years off the approval timeline.

Investor-grade risk pricing

PSI's MCMC engine runs 10,000 simulations of real ocean conditions to give lenders a probability-weighted IRR range — say 11–17% at 90% confidence — instead of a single analyst's guess.

Underwater data centers:

Same simulation architecture re-routed for a new use case, thermal plume impact, cooling intake/discharge, deep-water cable routing. Microsoft already proved the concept. PSI makes it rigorous.

DATA SOURCES:

- <https://data.marine.copernicus.eu/>
- <https://cds.climate.copernicus.eu/>
- <https://www.hycom.org/dataserver>
- <https://cioosatantic.ca/erddap/>



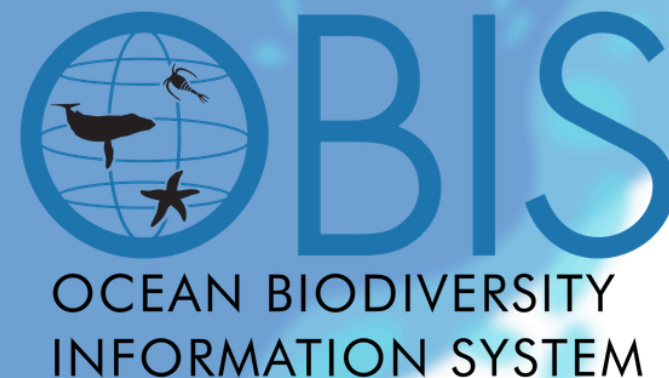
- <https://www.cnsopb.ns.ca/>
- <https://tidesandcurrents.noaa.gov/api/>
- <https://data.oceannetworks.ca/>
- <https://oceantrackingnetwork.org/>
- <https://members.oceantrack.org/erddap/>



DATA SOURCES:

- <https://www.nrcan.gc.ca/>
- <https://eccc-msc.github.io/open-data/>
- <https://www.smartatlantic.ca/>
- <https://obis.org/>
- <https://www.gebco.net/>

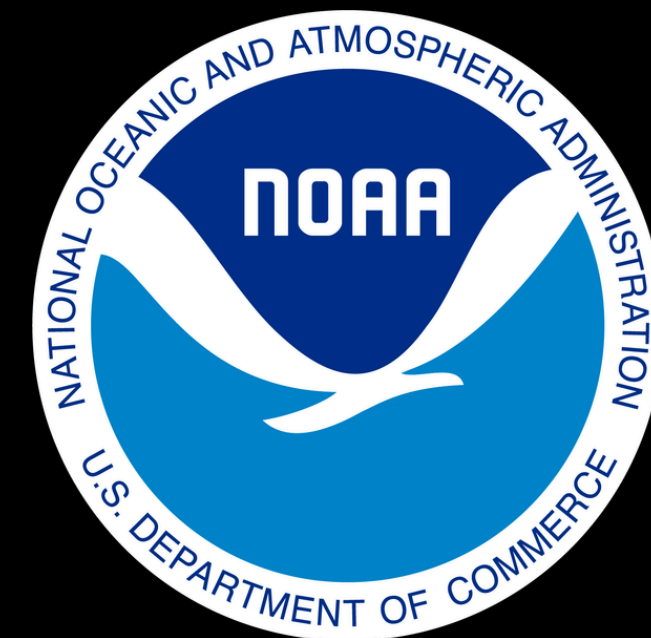
smartatlantic



Global Fishing Watch



- <https://globalfishingwatch.org/>
- <https://open-meteo.com/>
- <https://www.dwd.de/>
- <https://www.noaa.gov/>



Thank You!